K. Li, L. Yu, S. Wang, and P.-A. Heng, "Towards Cross-modality Medical Image Segmentation with Online Mutual Knowledge Distillation," In AAAI, 2020.

2020.3.19
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出发点

- 采用多种模态医学图像检测
- 不同模态具有不同的信息

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通过一个模态上的先验知识提高另一个模态上的分割性能

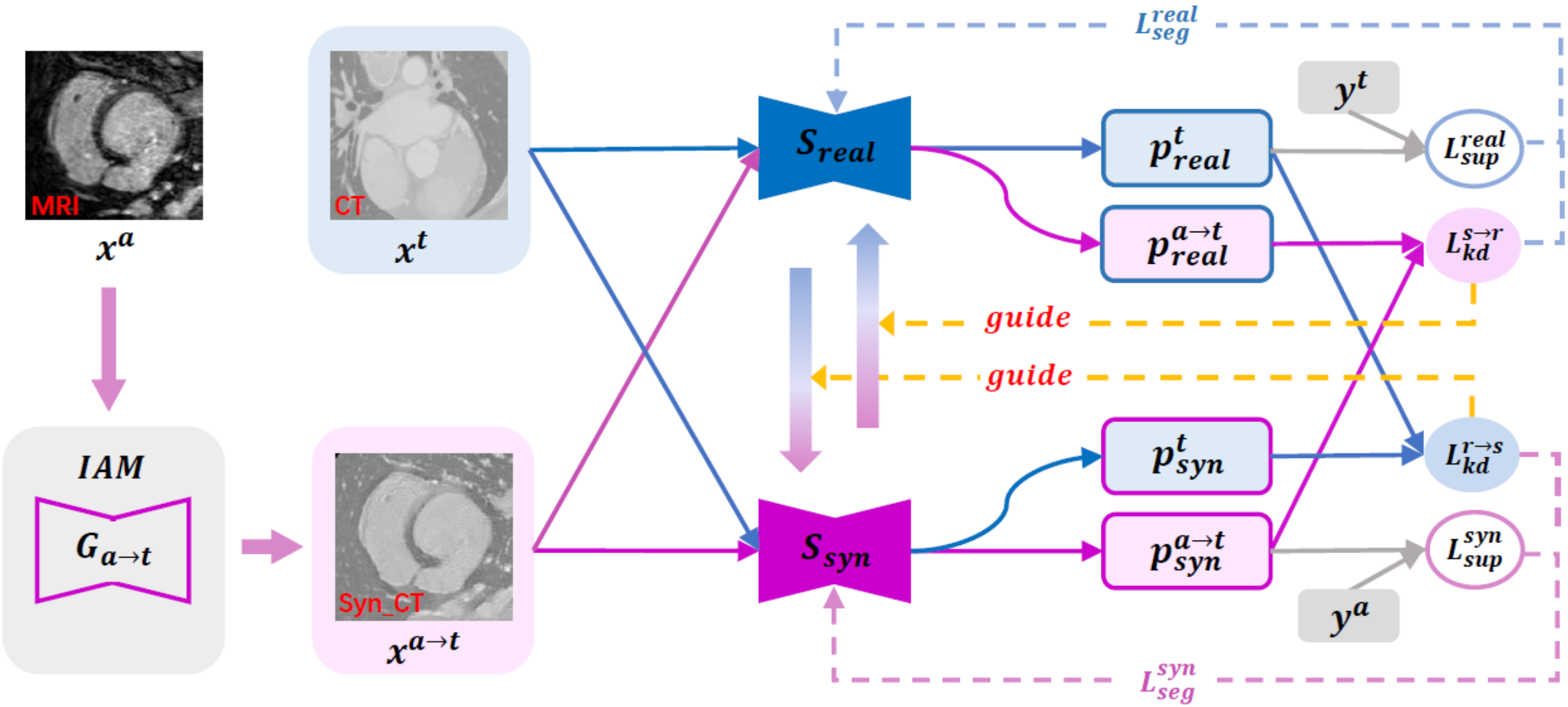
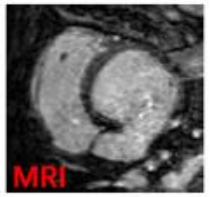
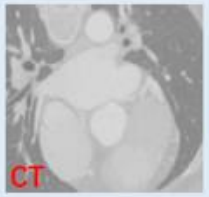


Figure 1: Overview of our framework, where magenta and blue represent the data flow of assistant modality (*e.g.*, MRI) and target modality (*e.g.*, CT) respectively. The generator $G_{a\to t}$ performs assistant-to-target translation and outputs synthesized CT data. The magenta-to-blue transition arrow represents the knowledge transfer from S_{syn} to S_{real} guided by $L_{kd}^{s\to r}$, while the blue-to-magenta transition arrow demonstrates the knowledge transfer from S_{real} to S_{syn} guided by $L_{kd}^{r\to s}$. In this way, both segmentors are mutually guided. In testing, we feed x_t into both segmentors and adopt the ensemble prediction as final results.

Image Alignment Module (IAM)



x^a



x^t

$$\mathcal{L}_{\text{adv}}^{t}(G_{a\to t}, D_t) = \mathbb{E}_{x^t\sim X^t}[\log D_t(x^t)] + \mathbb{E}_{x^a\sim X^a}[\log(1 - D_t(G_{a\to t}(x^a)))], \quad (1)$$

$$\mathcal{L}_{\text{adv}}^{a}(G_{t\to a}, D_a) = \mathbb{E}_{x^a\sim X^a}[\log D_a(x^a)] + \mathbb{E}_{x^t\sim X^t}[\log(1 - D_a(G_{t\to a}(x^t)))]. \quad (2)$$

$$\mathcal{L}_{cyc}(G_{a\to t}, G_{t\to a}) = \mathbb{E}_{x^a\sim X^a}\left\|\hat{x}^a - x^a\right\|_1 + \mathbb{E}_{x^t\sim X^t}\left\|\hat{x}^t - x^t\right\|_1. \quad (3)$$

$$\hat{x}^t = G_{a\to t}\left(G_{t\to a}\left(x^t\right)\right)$$

$$\mathcal{L}_{gan}^{a\to t} = \mathcal{L}_{adv}^{t} + \lambda_{cyc}\mathcal{L}_{cyc}, \quad (4)$$

$$\mathcal{L}_{gan}^{t\to a} = \mathcal{L}_{adv}^{a} + \lambda_{cyc}\mathcal{L}_{cyc}, \quad (5)$$

Mutual Knowledge Distillation

$$
\begin{aligned}
p_{syn}^{a\to t} &= S_{syn}\left(x^{a\to t}\right), \\
p_{real}^{t} &= S_{real}\left(x^{t}\right), \\
\mathcal{L}_{\text{sup}}^{\text{syn}} &= \mathcal{L}_{\text{ce}}\left(y^{a}, p_{syn}^{a\to t}\right) + \mathcal{L}_{\text{dice}}\left(y^{a}, p_{syn}^{a\to t}\right), \\
\mathcal{L}_{\text{sup}}^{\text{real}} &= \mathcal{L}_{\text{ce}}\left(y^{t}, p_{real}^{t}\right) + \mathcal{L}_{\text{dice}}\left(y^{t}, p_{real}^{t}\right),
\end{aligned}
\tag{6}
$$

$$
\mathcal{L}_{obj} = \mathcal{L}_{gan}^{a\to t} + \mathcal{L}_{gan}^{t\to a} + \mathcal{L}_{seg}^{syn} + \mathcal{L}_{seg}^{real}. \tag{9}
$$

Online Mutual Training

Algorithm 1 Training procedure of the proposed approach

Input: A batch of (x^t, y^t) from target-modality dataset X^t and (x^a, y^a) from assistant-modality dataset X^a

Output: The prediction p^t of input x^t

$\theta_{G_{a\to t}}, \theta_{G_{t\to a}}, \theta_{D_a}, \theta_{D_t}, \theta_{S_{syn}}, \theta_{S_{real}} \leftarrow$ initialize

while not converge **do**

  $(x^t, y^t), (x^a, y^a) \leftarrow$ sampled from X^t and X^a

  Calculate $\mathcal{L}_{gan}^{a\to t}$ and $\mathcal{L}_{sup}^{syn}$ as Eq. (4) and (6)

  Update $\theta_{G_{a\to t}} \overset{+}{\leftarrow} -\Delta_{\theta_{G_{a\to t}}} \left(\mathcal{L}_{gan}^{a\to t} + \mathcal{L}_{sup}^{syn}\right)$

  Calculate $\mathcal{L}_{adv}^{t}$ as Eq. (1)

  Update $\theta_{D_t} \overset{+}{\leftarrow} -\Delta_{\theta_{D_t}} \mathcal{L}_{adv}^{t}$

  Calculate $\mathcal{L}_{gan}^{t\to a}$ as Eq. (5)

  Update $\theta_{G_{t\to a}} \overset{+}{\leftarrow} -\Delta_{\theta_{G_{t\to a}}} \mathcal{L}_{gan}^{t\to a}$

  Calculate $\mathcal{L}_{adv}^{a}$ as Eq. (2)

  Update $\theta_{D_a} \overset{+}{\leftarrow} -\Delta_{\theta_{D_a}} \mathcal{L}_{adv}^{a}$

  Calculate $\mathcal{L}_{seg}^{syn}$ and $\mathcal{L}_{seg}^{real}$ as Eq. (8)

  Update $\theta_{S_{syn}} \overset{+}{\leftarrow} -\Delta_{\theta_{S_{syn}}} \mathcal{L}_{seg}^{syn}$

  Update $\theta_{S_{real}} \overset{+}{\leftarrow} -\Delta_{\theta_{S_{real}}} \mathcal{L}_{seg}^{real}$

Calculate p_{real}^t and p_{syn}^t

return $p^t \leftarrow$ the ensemble of p_{real}^t and p_{syn}^t

Table 1: Quantitative comparison between our method and other multi-modality segmentation methods. Here we take CT as target modality and MR as assistant modality. The dice of all heart substructures and the average of them are reported here.

Method	Mean Dice	Dice of substructures of heart						
		MYO	LA	LV	RA	RV	AA	PA
Baseline	0.8706	0.8702	0.8922	0.9086	0.8386	0.846	0.9252	0.8134
Fine-tune	0.8769	0.8716	0.9040	0.9079	0.8443	0.8526	0.9274	0.8305
Joint-training	0.8743	0.8665	0.9076	0.9123	0.8278	0.8492	0.9302	0.8266
X-shape (2018)	0.8767	0.8719	0.8979	0.9094	0.8551	0.8444	0.9343	0.8240
Jiang et al. (2018)	0.8765	0.8723	0.9054	0.9073	0.8338	0.8525	0.9484	0.8156
Zhang et al. (2018b)	0.8850	0.8781	0.9112	0.9134	0.8514	0.8631	0.9430	0.8342
Ours (S_{syn})	0.8849	0.8738	0.9086	0.9169	0.8571	0.8633	0.9412	0.8333
Ours (S_{real})	0.8947	0.8893	0.9131	0.9197	0.8668	0.8758	0.9552	0.8432
Ours	**0.9012**	**0.8934**	**0.9190**	**0.9267**	**0.8747**	**0.8814**	**0.9595**	**0.8538**

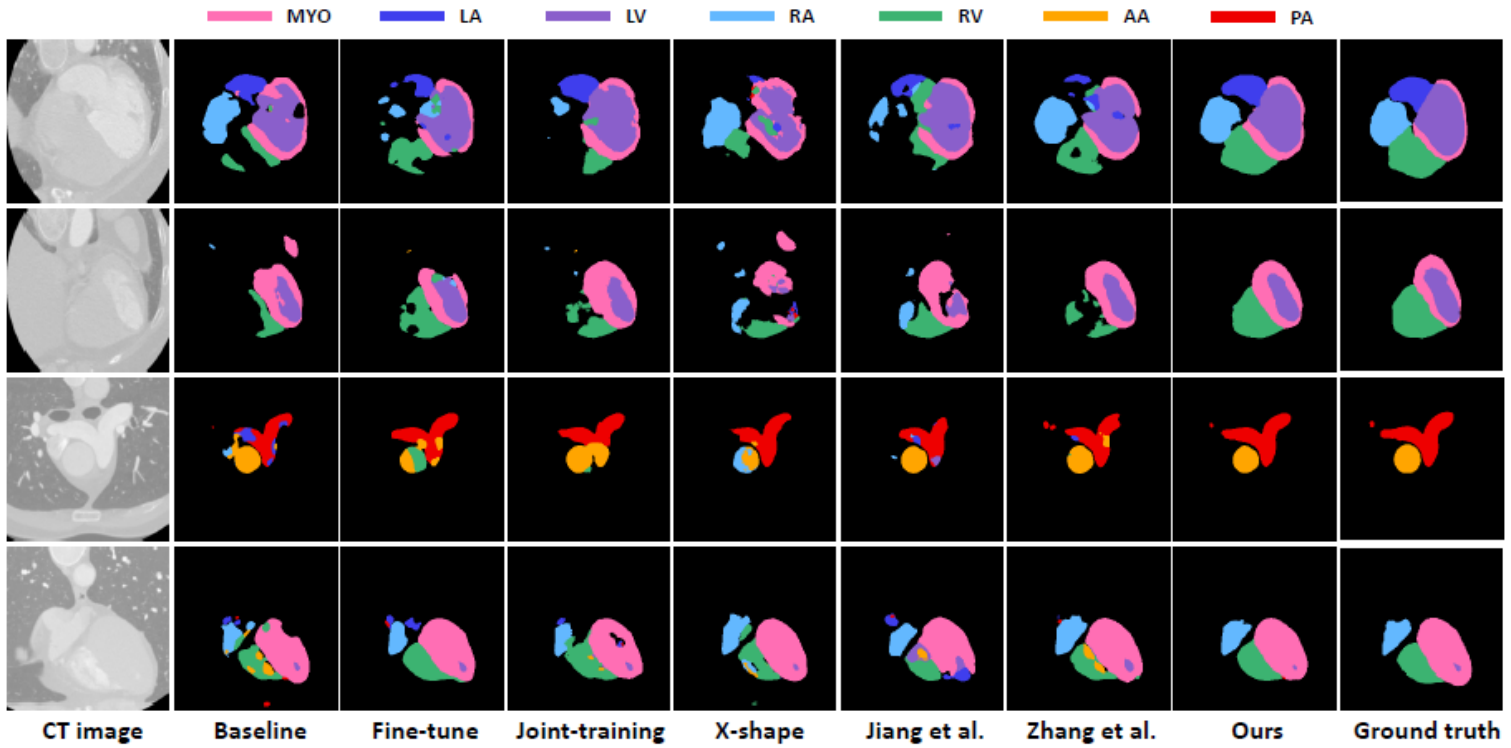


Figure 2: Visual comparison of segmentation results produced by different methods, where the legend for cardiac substructures

实验

Table 2: Ablation study of key components in our method, where the mean dice of all heart substructures is reported.

Methods	S_{syn}	S_{real}	S_{ens}
W/o IAM	0.8730	0.8745	0.8829
W/o $\mathcal{L}_{kd}^{s\rightarrow r}$	0.8825	0.8742	0.8879
W/o $\mathcal{L}_{kd}^{r\rightarrow s}$	0.5785	0.8904	0.7937
Ours	**0.8849**	**0.8947**	**0.9012**

Table 3: Performance comparison of different amounts of assisted modality data in training.

Setting	Mean Dice
5-MRI	0.8916
10-MRI	0.8945
20-MRI	**0.9012**

实验

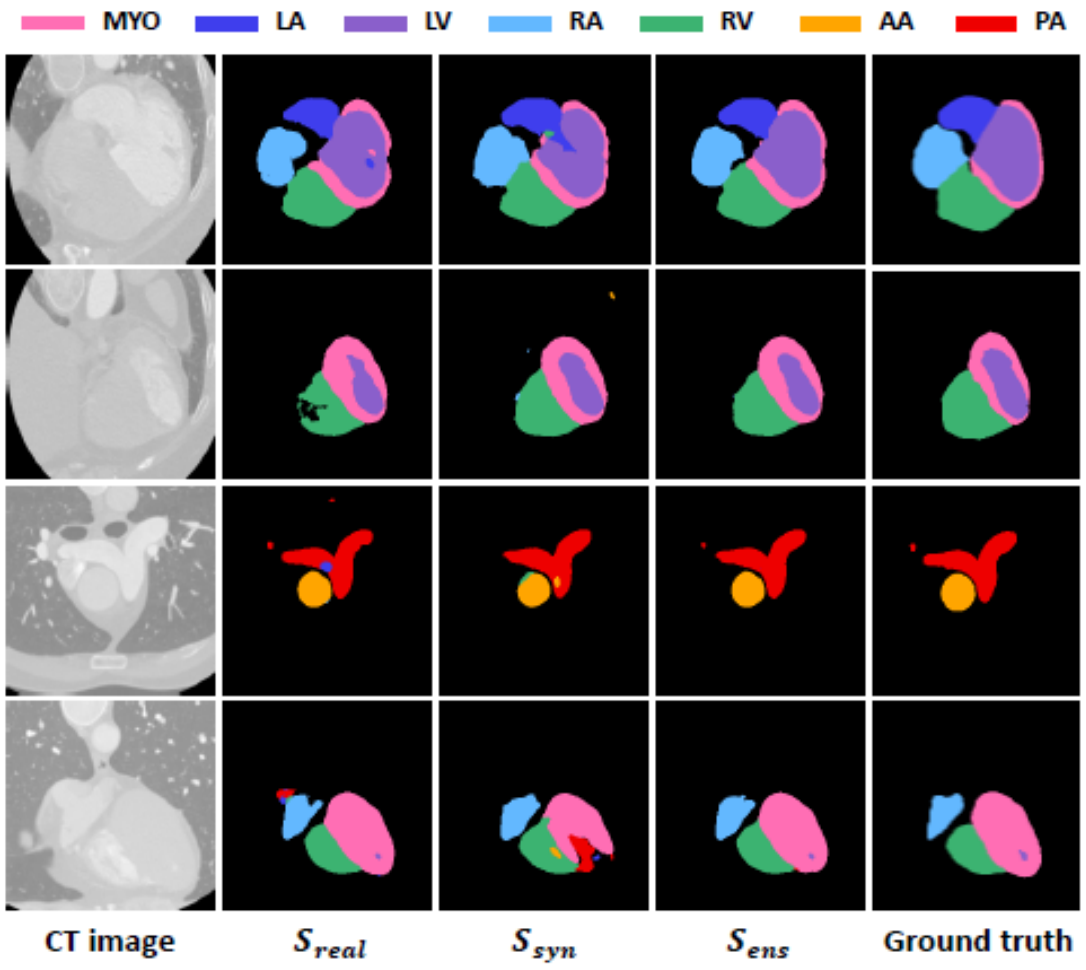


Figure 4: Visual comparison of segmentation results by S_{real}, S_{syn} and the ensemble segmentor, which could balance inconsistent predictions and produce reliable results.

实验

Table 4: Quantitative comparison between our approach and other multi-modality segmentation methods in whole heart segmentation. Here we take MR as target modality and CT as assistant modality. The dice of all cardiac substructures and mean dice are reported.

Method	Mean Dice	Dice of substructures of heart						
		MYO	LA	LV	RA	RV	AA	PA
Baseline	0.8243	0.8259	0.8153	0.9347	0.8146	0.9140	0.7365	0.7288
Fine-tune	0.8217	0.8336	0.8447	0.9371	0.7893	0.8940	0.7015	0.7516
Joint-training	0.8306	0.8375	0.7893	0.9303	0.8507	0.9119	0.7487	0.7455
X-shape (2018)	0.8337	0.8321	0.8247	0.9347	0.8450	0.9181	0.7278	0.7536
Jiang et al. (2018)	0.8370	0.8309	0.8569	0.9356	0.8582	0.9016	0.7334	0.7422
Zhang et al. (2018b)	0.8375	0.8360	0.8169	0.9317	0.8501	**0.9183**	0.7435	**0.7663**
Ours (S_{syn})	0.8382	0.8405	0.8308	0.9337	**0.8655**	0.9136	0.7351	0.7483
Ours (S_{real})	0.8466	0.8507	0.8582	0.9427	0.8623	0.9091	0.7398	0.7633
Ours	**0.8517**	**0.8525**	**0.8695**	**0.9437**	**0.8655**	0.9121	**0.7544**	0.7640

Thank You!